

# Conjectures on Witt Cohomology of Stabilizer Codes

A Formal Framework for Dieudonné-Module Theory in  
Quantum Error Correction

QNFO Research Agent — Project: `number-theory-ultrametric-deep`

2026-07-03 — Status: Phase 1 — Framework Formalization

## Abstract

This document formalizes the Witt cohomology conjecture (Pillar VI of the number–QEC bridge program) and its cross-pillar connections to Fontaine’s  $p$ -adic Hodge theory (Pillar III). We define a cohomology theory for stabilizer codes valued in modules over the Dieudonné ring, identify the Frobenius and Verschiebung operators with Clifford conjugation and  $T$ -gate injection, and state the Dieudonné–Manin slope-threshold correspondence as a precise inequality. Every conjecture is labeled with its certainty status and includes an explicit falsifiability condition.

## Contents

|          |                                                               |          |
|----------|---------------------------------------------------------------|----------|
| <b>1</b> | <b>Preliminaries on Witt Vectors and Dieudonné Modules</b>    | <b>2</b> |
| 1.1      | Witt Vectors . . . . .                                        | 2        |
| 1.2      | The Dieudonné Ring . . . . .                                  | 3        |
| 1.3      | The Dieudonné–Manin Slope Decomposition . . . . .             | 3        |
| 1.4      | Fontaine’s Period Rings . . . . .                             | 4        |
| <b>2</b> | <b>Formal Definition: Witt Cohomology of Stabilizer Codes</b> | <b>4</b> |
| 2.1      | Stabilizer Codes over Finite Fields . . . . .                 | 4        |
| 2.2      | The Witt Cohomology Functor . . . . .                         | 4        |
| 2.3      | The Frobenius Operator: Clifford Conjugation . . . . .        | 5        |
| 2.4      | The Verschiebung Operator: $T$ -Gate Injection . . . . .      | 6        |
| 2.5      | The Witt-Cohomology Pairing . . . . .                         | 6        |
| <b>3</b> | <b>Dieudonné–Manin Classification of Codes</b>                | <b>6</b> |
| 3.1      | Slopes of a Stabilizer Code . . . . .                         | 6        |
| 3.2      | Crystalline and Non-Crystalline Codes . . . . .               | 7        |
| 3.3      | Slope Semantics and Error Correction . . . . .                | 8        |

|          |                                                            |           |
|----------|------------------------------------------------------------|-----------|
| <b>4</b> | <b>Witt Vectors as Truncated Hierarchies</b>               | <b>8</b>  |
| 4.1      | Truncation and Code Levels . . . . .                       | 8         |
| 4.2      | Thermodynamic Limit . . . . .                              | 9         |
| 4.3      | Connection to Moy–Prasad Depth . . . . .                   | 9         |
| <b>5</b> | <b>The Fontaine Classification (Cross-Pillar)</b>          | <b>9</b>  |
| 5.1      | The Code Filtration . . . . .                              | 9         |
| 5.2      | Interpretation of Fontaine Classes . . . . .               | 10        |
| 5.3      | The Frobenius–Cyclic Measurement Correspondence . . . . .  | 10        |
| 5.4      | The Crystalline–Topological Link . . . . .                 | 11        |
| <b>6</b> | <b>Computational Consequences</b>                          | <b>11</b> |
| 6.1      | Immediately Testable Predictions . . . . .                 | 11        |
| 6.2      | Predictions Requiring Deeper Theory . . . . .              | 12        |
| 6.3      | Falsifiability of the Entire Pillar VI Framework . . . . . | 12        |
| 6.4      | Computational Implementation Roadmap . . . . .             | 13        |

# 1 Preliminaries on Witt Vectors and Dieudonné Modules

We recall the classical theory needed to formulate the quantum-code analog. All material in this section is [\[established\]](#).

## 1.1 Witt Vectors

Let  $p$  be a prime and let  $k$  be a perfect field of characteristic  $p$ . Let  $q = p^m$  for  $m \geq 1$ .

**Definition 1.1** ( $p$ -typical Witt vectors). The *ring of ( $p$ -typical) Witt vectors*  $W(k)$  is the set of infinite sequences

$$W(k) = \{(a_0, a_1, a_2, \dots) : a_i \in k\}$$

equipped with addition and multiplication defined by the universal Witt polynomials

$$\Phi_n(x_0, \dots, x_{n-1}) = \sum_{i=0}^{n-1} p^i x_i^{p^{n-i}}, \quad P_n(x_0, \dots, x_{n-1}; y_0, \dots, y_{n-1})$$

such that the *ghost map*

$$w : W(k) \longrightarrow k^{\mathbb{N}}, \quad w(a) = (\Phi_n(a_0, \dots, a_{n-1}))_{n \geq 1}$$

is a ring homomorphism. The ring  $W(k)$  is a complete discrete valuation ring of characteristic 0 with maximal ideal  $pW(k)$  and residue field  $k$ .

**Definition 1.2** (Truncated Witt vectors). For  $n \geq 1$ , the *ring of truncated Witt vectors of length  $n$*  is

$$W_n(k) = W(k)/p^n W(k) \cong \{(a_0, \dots, a_{n-1}) : a_i \in k\}$$

with the operations induced from  $W(k)$ . The canonical projection is

$$R_{n+1} : W_{n+1}(k) \twoheadrightarrow W_n(k), \quad R_{n+1}(a_0, \dots, a_n) = (a_0, \dots, a_{n-1}).$$

The full Witt ring is the inverse limit:

$$W(k) = \varprojlim_n W_n(k).$$

**Definition 1.3** (Teichmüller lift). The *Teichmüller representative* of  $a \in k$  is

$$[a] = (a, 0, 0, \dots) \in W(k).$$

The map  $a \mapsto [a]$  is a multiplicative section of the reduction map  $W(k) \rightarrow k$ .

## 1.2 The Dieudonné Ring

**Definition 1.4** (Dieudonné ring). Let  $k$  be a perfect field of characteristic  $p$ . The *Dieudonné ring*  $D_k$  is the non-commutative  $W(k)$ -algebra generated by two operators  $F$  and  $V$  subject to the relations

$$FV = VF = p, \quad Fa = \sigma(a)F, \quad Va = \sigma^{-1}(a)V \quad (\forall a \in W(k)),$$

where  $\sigma : W(k) \rightarrow W(k)$  is the Frobenius automorphism induced by  $x \mapsto x^p$  on the residue field  $k$ . Explicitly,

$$D_k = W(k)\langle F, V \rangle / (FV - p, VF - p).$$

**Definition 1.5** (Dieudonné module). A *Dieudonné module* over  $k$  is a left  $D_k$ -module  $M$  that is finitely generated and free as a  $W(k)$ -module. The category of Dieudonné modules is denoted  $D\text{-}\mathbf{Mod}_k$ .

On a Dieudonné module  $M$ , the operator  $F$  is  $\sigma$ -semilinear and  $V$  is  $\sigma^{-1}$ -semilinear. The relation  $FV = VF = p$  implies  $pM \subseteq F(M) \cap V(M)$ .

## 1.3 The Dieudonné–Manin Slope Decomposition

**Proposition 1.1** (Dieudonné–Manin classification, [\[established\]](#)). *Let  $M$  be a Dieudonné module over  $k$ . Then the isocrystal  $M \otimes_{W(k)} \text{Frac}(W(k))$  admits a canonical direct-sum decomposition into slope components:*

$$M \otimes_{\mathbb{Z}_p} \mathbb{Q}_p \cong \bigoplus_{\lambda \in \mathbb{Q}} M_\lambda,$$

where each  $M_\lambda$  is an isoclinic module of slope  $\lambda = r/s$  (with  $\gcd(r, s) = 1$ ,  $s > 0$ ), characterized by the existence of a  $W(k)$ -lattice  $L \subseteq M_\lambda$  such that

$$F^s(L) = p^r L.$$

The multiplicity of slope  $\lambda$  is the integer  $s \cdot \dim_{\text{Frac}(W(k))} M_\lambda$ .

*Remark 1.1.* The slope  $\lambda = 0$  corresponds to *étale* modules (pure Frobenius-action),  $\lambda = 1$  to *multiplicative type* (pure Verschiebung-action), and  $\lambda = 1/2$  to *self-dual* modules.

## 1.4 Fontaine's Period Rings

We recall the definitions needed for the cross-pillar Fontaine classification (§5).

**Definition 1.6** (Fontaine's period rings, [\[established\]](#)). For a finite extension  $K/\mathbb{Q}_p$ , Fontaine constructs the following  $\mathbb{Q}_p$ -algebras equipped with a continuous  $G_K$ -action:

$$B_{\text{crys}} \subset B_{\text{st}} \subset B_{\text{dR}},$$

where:

- $B_{\text{dR}}$  is the *de Rham period ring* — a complete discrete valuation field with residue field  $\mathbb{C}_p$ , endowed with the de Rham filtration  $\text{Fil}^\bullet B_{\text{dR}}$ .
- $B_{\text{st}} = B_{\text{crys}}[\log t]$  is the *semistable period ring*, where  $t$  is Fontaine's  $2\pi i$  element, equipped with a monodromy operator  $N$ .
- $B_{\text{crys}}$  is the *crystalline period ring*, the maximal subring of  $B_{\text{st}}$  on which  $N = 0$ , equipped with a  $\sigma$ -semilinear Frobenius  $\varphi : B_{\text{crys}} \rightarrow B_{\text{crys}}$ .
- $B_{\text{HT}} = \text{gr}^\bullet B_{\text{dR}}$  is the *Hodge-Tate period ring*, the associated graded ring.

**Definition 1.7** (Crystalline, semistable, de Rham representations). A  $p$ -adic Galois representation  $V$  of  $G_K$  is:

- *de Rham* if  $\dim_K(V \otimes_{\mathbb{Q}_p} B_{\text{dR}})^{G_K} = \dim_{\mathbb{Q}_p} V$ ,
- *semistable* if  $\dim_K(V \otimes_{\mathbb{Q}_p} B_{\text{st}})^{G_K} = \dim_{\mathbb{Q}_p} V$ ,
- *crystalline* if  $\dim_K(V \otimes_{\mathbb{Q}_p} B_{\text{crys}})^{G_K} = \dim_{\mathbb{Q}_p} V$ .

We have the strict inclusions:

$$\{\text{crystalline}\} \subset \{\text{semistable}\} \subset \{\text{de Rham}\}.$$

## 2 Formal Definition: Witt Cohomology of Stabilizer Codes

### 2.1 Stabilizer Codes over Finite Fields

We fix a ground field  $k = \mathbb{F}_q$  where  $q = p^m$ .

**Definition 2.1** (Stabilizer code over  $k$ ). A *stabilizer code*  $\mathcal{C}$  over  $k$  with parameters  $[[n, \kappa, \delta]]_q$  is a  $q^\kappa$ -dimensional subspace of the Hilbert space  $(\mathbb{C}^q)^{\otimes n}$  whose stabilizer group  $S \subseteq \mathcal{P}_q^n$  is an abelian subgroup of the generalized  $q$ -dimensional Pauli group that does not contain  $-I$ . The code corrects errors on up to  $\lfloor (\delta - 1)/2 \rfloor$  qudits.

We denote by  $\mathfrak{C}_p$  the set of isomorphism classes of stabilizer codes defined over finite fields of characteristic  $p$ .

### 2.2 The Witt Cohomology Functor

**Conjecture 2.1** (Existence of Witt cohomology, [\[my conjecture\]](#)). *There exists a contravariant functor*

$$H_{\text{stab}}^\bullet : \mathfrak{C}_p \longrightarrow D\text{-Mod}_{\mathbb{F}_p}, \quad \mathcal{C} \longmapsto H_{\text{stab}}^\bullet(\mathcal{C}),$$

assigning to every stabilizer code  $\mathcal{C}$  over  $\mathbb{F}_q$  (with  $q = p^m$ ) a  $\mathbb{Z}$ -graded Dieudonné module  $H_{\text{stab}}^\bullet(\mathcal{C}) = \bigoplus_{i \geq 0} H_{\text{stab}}^i(\mathcal{C})$  over  $W(\mathbb{F}_p)$  with the following properties:

- (1) **Functoriality:** A morphism of stabilizer codes (a  $k$ -linear map of the underlying stabilizer groups that preserves the code subspace) induces a morphism of Dieudonné modules in the reverse direction.
- (2) **Finite generation:** Each  $H_{\text{stab}}^i(\mathcal{C})$  is a finitely generated  $W(\mathbb{F}_p)$ -module.
- (3) **Code dimension:**  $\dim_{\mathbb{Q}_p} H_{\text{stab}}^0(\mathcal{C}) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p = \kappa$  (dimension of the encoded logical subspace).
- (4) **Distance grading:**  $\text{rk}_{W(\mathbb{F}_p)} H_{\text{stab}}^i(\mathcal{C}) = 0$  for  $i > \delta$  (cohomology vanishes beyond the code distance).
- (5) **Mayer–Vietoris:** For a covering of the stabilizer by two commuting subgroups  $S = S_1 S_2$  with  $S_{12} = S_1 \cap S_2$ , there is a long exact sequence of Dieudonné modules related to the induced codes  $\mathcal{C}(S_1), \mathcal{C}(S_2), \mathcal{C}(S_{12})$ .

*Remark 2.1* (Falsifiability of Conjecture 2.1). The conjecture would be falsified if no non-trivial  $D_k$ -module structure can be defined on any cohomological invariant of stabilizer codes, or if the functoriality condition (1) forces  $H_{\text{stab}}^\bullet(\mathcal{C})$  to be the zero module for all  $\mathcal{C}$ .

## 2.3 The Frobenius Operator: Clifford Conjugation

**Definition 2.2** (Clifford conjugation of a stabilizer code). Let  $\mathcal{C}$  be a stabilizer code with stabilizer group  $S$ . The *Clifford conjugate* of  $\mathcal{C}$  is the code

$$\mathcal{C}^F = U \mathcal{C} U^\dagger,$$

where  $U$  is the full Clifford-group element that conjugates the generalized Pauli group by a complete Fourier transform on each qudit. The stabilizer group of  $\mathcal{C}^F$  is the image of  $S$  under this conjugation.

**Conjecture 2.2** (Frobenius as Clifford conjugation, [\[my conjecture\]](#)). Under the Witt cohomology functor of Conjecture 2.1, the Frobenius operator

$$F : H_{\text{stab}}^\bullet(\mathcal{C}) \longrightarrow H_{\text{stab}}^\bullet(\mathcal{C}^F)$$

corresponds to the  $\sigma$ -semilinear map induced by Clifford conjugation. Moreover,  $F$  is an isomorphism on the isocrystal level:

$$F \otimes \mathbb{Q}_p : H_{\text{stab}}^\bullet(\mathcal{C}) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p \xrightarrow{\cong} H_{\text{stab}}^\bullet(\mathcal{C}^F) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p.$$

*Remark 2.2* (Falsifiability of Conjecture 2.2). The conjecture is falsified if, for a known code family (e.g., the  $[[5, 1, 3]]$  perfect code or toric codes), the iterated Clifford conjugate  $\mathcal{C}^{F^k}$  has cohomological invariants that do not follow the slope pattern predicted by the Frobenius action on a Dieudonné module.

## 2.4 The Verschiebung Operator: $T$ -Gate Injection

**Definition 2.3** ( $T$ -gate injection (level increase)). Let  $\mathcal{C}$  be a stabilizer code over  $\mathbb{F}_p$ . The  $T$ -gate injected code  $\mathcal{C}^V$  is obtained by conjugating the stabilizer generators of  $\mathcal{C}$  by a  $T$ -gate ( $\pi/8$  phase gate) on a designated subset of qudits:

$$\mathcal{C}^V = T \mathcal{C} T^\dagger,$$

where  $T$  acts as  $T|j\rangle = e^{2\pi i j^2/8p}|j\rangle$  on qudits of dimension  $q = p^m$ . The operator  $V$  encodes the effect of *increasing the Clifford hierarchy level* by one — it “unfolds” the code into the next level of the hierarchy, corresponding to the shift action of Verschiebung on Witt vector components.

**Conjecture 2.3** (Verschiebung as  $T$ -gate injection, [my conjecture]). *Under the Witt cohomology functor, the Verschiebung operator*

$$V : H_{\text{stab}}^\bullet(\mathcal{C}^V) \longrightarrow H_{\text{stab}}^\bullet(\mathcal{C})$$

*is a  $\sigma^{-1}$ -semilinear map corresponding to the inverse of  $T$ -gate injection at the cohomological level. Moreover, on the underlying Dieudonné module:*

$$FV = p = VF \quad \text{on } H_{\text{stab}}^\bullet(\mathcal{C}).$$

*Remark 2.3* (Falsifiability of Conjecture 2.3). The conjecture is falsified if, for any stabilizer code  $\mathcal{C}$ , the composition of Clifford conjugation followed by  $T$ -gate injection (or vice versa) fails to equal the scalar  $p$  on the available cohomological invariants. A direct numerical test: compute the trace of  $F \circ V$  and  $V \circ F$  on  $H_{\text{stab}}^0$  for small codes and compare to  $p \cdot \text{rk}_{W(\mathbb{F}_p)} H_{\text{stab}}^0(\mathcal{C})$ .

## 2.5 The Witt-Cohomology Pairing

**Conjecture 2.4** (Poincaré duality for stabilizer codes, [speculative]). *For a stabilizer code  $\mathcal{C}$  with parameters  $[[n, \kappa, \delta]]_q$ , there exists a non-degenerate pairing*

$$\langle \cdot, \cdot \rangle : H_{\text{stab}}^i(\mathcal{C}) \times H_{\text{stab}}^{\delta-i}(\mathcal{C}^\perp) \longrightarrow W(\mathbb{F}_p),$$

*where  $\mathcal{C}^\perp$  is the dual code. Under this pairing,  $F$  and  $V$  are adjoint:*

$$\langle Fx, y \rangle = \langle x, Vy \rangle^\sigma.$$

# 3 Dieudonné–Manin Classification of Codes

## 3.1 Slopes of a Stabilizer Code

**Definition 3.1** (Slope spectrum). For a stabilizer code  $\mathcal{C}$ , its *slope spectrum* is the multiset of rational numbers

$$\text{Slopes}(\mathcal{C}) = \{\lambda_1, \dots, \lambda_r\} \subset \mathbb{Q} \cap [0, 1]$$

obtained from the Dieudonné–Manin decomposition of the isocrystal  $H_{\text{stab}}^\bullet(\mathcal{C}) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p$ :

$$H_{\text{stab}}^\bullet(\mathcal{C}) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p \cong \bigoplus_{\lambda \in \text{Slopes}(\mathcal{C})} M_\lambda,$$

where  $M_\lambda$  is isoclinic of slope  $\lambda$ . The *slope spread* is  $\Delta(\mathcal{C}) = \lambda_{\max} - \lambda_{\min}$ .

**Conjecture 3.1** (Slope-threshold inequality, [\[my conjecture\]](#)). *Let  $\mathcal{C}$  be a family of stabilizer codes  $\{\mathcal{C}_n\}_{n=1}^\infty$  with an associated fault-tolerance threshold  $p_{\text{th}}(\mathcal{C})$ . There exists a universal monotone-decreasing function  $f : [0, 1] \rightarrow [0, 1]$  such that*

$$p_{\text{th}}(\mathcal{C}) \leq f(\Delta(\mathcal{C}))$$

for every code family  $\mathcal{C}$ , with equality holding for crystalline code families (those with integer slopes; see Definition 3.2). Moreover:

- $\lambda = 0$  (purely étale)  $\implies p_{\text{th}} = 1$  (perfect, no logical error in the thermodynamic limit),
- $\lambda = 1$  (multiplicative)  $\implies p_{\text{th}} = 0$  (classical, no entanglement protection),
- $\lambda = 1/2$  (self-dual)  $\implies p_{\text{th}} = 1/2$  (CSS codes with  $H_X = H_Z$ ).

*Remark 3.1* (Falsifiability of Conjecture 3.1). This conjecture is directly testable: compute the slope spectrum of known code families (surface codes, color codes, toric codes, quantum LDPC codes) and correlate  $\Delta(\mathcal{C})$  with their independently determined error thresholds. No correlation between  $\Delta$  and  $p_{\text{th}}$  (after controlling for  $n, k, d$ ) would falsify the conjecture. For code families with a single slope  $\lambda$ , the prediction  $p_{\text{th}} = 1 - \lambda$  is a sharp numerical test.

## 3.2 Crystalline and Non-Crystalline Codes

**Definition 3.2** (Crystalline code, [\[my conjecture\]](#)). A stabilizer code  $\mathcal{C}$  is *crystalline* if all slopes in its slope spectrum are integers (hence 0 or 1, since slopes are confined to  $[0, 1]$ ). Equivalently, the Frobenius operator  $F$  is invertible on  $H_{\text{stab}}^\bullet(\mathcal{C})$  (no non-trivial  $V$ -torsion).

**Conjecture 3.2** (Characterization of crystalline codes, [\[my conjecture\]](#)). *A stabilizer code  $\mathcal{C}$  is crystalline if and only if its stabilizer group is invariant under the full Clifford group (Frobenius-invariant):*

$$USU^\dagger = S \quad \forall U \in \mathcal{C}\ell_n^{(q)}.$$

*Such codes are topological in the sense that their error-correction properties are invariant under local unitary deformations.*

*Remark 3.2* (Falsifiability of Conjecture 3.2). Test on known topological codes: the toric code and color code are expected to be crystalline (all slopes 0); generic random stabilizer codes should have fractional slopes. If a known topological code is not crystalline, or if a known non-topological code is crystalline, the conjecture is falsified.

### 3.3 Slope Semantics and Error Correction

**Proposition 3.1** (Slope semantics, [my conjecture]). *The slope  $\lambda$  of a Dieudonné-module component  $M_\lambda \subseteq H_{\text{stab}}^\bullet(\mathcal{C}) \otimes \mathbb{Q}_p$  has the following QEC interpretation:*

| Slope $\lambda$ | Number-theoretic type                       | QEC meaning                                               |
|-----------------|---------------------------------------------|-----------------------------------------------------------|
| 0               | <i>Étale (pure <math>F</math>)</i>          | <i>Perfect fault-tolerance; exponential suppression</i>   |
| $(0, 1/2)$      | <i>Fractional, <math>F</math>-dominant</i>  | <i>Strong error correction; high threshold</i>            |
| $1/2$           | <i>Self-dual</i>                            | <i>CSS code with symmetric <math>X/Z</math> structure</i> |
| $(1/2, 1)$      | <i>Fractional, <math>V</math>-dominant</i>  | <i>Weak error correction; low threshold</i>               |
| 1               | <i>Multiplicative (pure <math>V</math>)</i> | <i>Classical code; no quantum advantage</i>               |
| $\notin [0, 1]$ | <i>Unphysical</i>                           | <i>No quantum code can realize these slopes</i>           |

## 4 Witt Vectors as Truncated Hierarchies

### 4.1 Truncation and Code Levels

**Definition 4.1** (Truncated code at level  $n$ ). For a stabilizer code  $\mathcal{C}$  over  $\mathbb{F}_q$ , its  $n$ th truncation is

$$\mathcal{C}_n = \mathcal{C} \otimes_{\mathbb{F}_q} W_n(\mathbb{F}_q),$$

interpreted as the code with stabilizer entries lifted to the Witt vectors of length  $n$ . The family  $\{\mathcal{C}_n\}_{n \geq 1}$  forms a projective system with reduction maps

$$\mathcal{C}_{n+1} \twoheadrightarrow \mathcal{C}_n,$$

induced by the Witt vector truncation  $W_{n+1}(\mathbb{F}_q) \twoheadrightarrow W_n(\mathbb{F}_q)$ .

**Conjecture 4.1** (Witt hierarchy correspondence, [my conjecture]). *The  $n$ th truncation  $\mathcal{C}_n$  corresponds to a quantum code with  $(n - 1)$ -st order error correction:*

| $W_n$ level                                       | QEC interpretation                                                 |
|---------------------------------------------------|--------------------------------------------------------------------|
| $W_1 = \mathbb{F}_q$                              | <i>Qubit/qutrit-level code (mod-<math>p</math> information)</i>    |
| $W_2$                                             | <i>Code with single-order syndrome correction</i>                  |
| $W_3$                                             | <i>Code with second-order syndrome correction</i>                  |
| $W_n$                                             | <i>Code with <math>(n - 1)</math>-st order syndrome correction</i> |
| $W(\mathbb{F}_q) = \varprojlim W_n(\mathbb{F}_q)$ | <b>Thermodynamic limit:</b> <i>infinite code family</i>            |

Moreover, properties that hold for all finite  $n$  but fail in the limit  $W(\mathbb{F}_q)$  indicate phase transitions in code space.

*Remark 4.1* (Falsifiability of Conjecture 4.1). Construct the truncated codes  $\mathcal{C}_n$  for a known family (e.g., concatenated codes or repetition codes) and verify that the  $n$ th level indeed provides  $(n - 1)$ -st order correction. If the truncation level does not correlate with error-correction order, the hierarchy interpretation is falsified.

## 4.2 Thermodynamic Limit

**Definition 4.2** (Thermodynamic limit of a code family). The *thermodynamic limit* of a code family  $\{\mathcal{C}^{(n)}\}$  is the inverse limit

$$\mathcal{C}^{(\infty)} = \varprojlim_n \mathcal{C}^{(n)},$$

taken in the category of stabilizer codes with reduction maps given by partial trace or concatenation decoding.

**Conjecture 4.2** (Witt limit = thermodynamic limit, [\[my conjecture\]](#)). *For a code family  $\mathcal{C}^{(n)}$  that arises from the Witt truncation  $\mathcal{C}_n = \mathcal{C} \otimes_{\mathbb{F}_q} W_n(\mathbb{F}_q)$ , the thermodynamic limit is isomorphic to the code over the full Witt vector ring:*

$$\mathcal{C}^{(\infty)} \cong \mathcal{C} \otimes_{\mathbb{F}_q} W(\mathbb{F}_q).$$

*The Dieudonné module of the limit code satisfies*

$$H_{\text{stab}}^\bullet(\mathcal{C}^{(\infty)}) \cong H_{\text{stab}}^\bullet(\mathcal{C}) \otimes_{W(\mathbb{F}_p)} W(\mathbb{F}_q),$$

*where the right-hand side is the extension of scalars to the larger Witt ring.*

## 4.3 Connection to Moy–Prasad Depth

**Conjecture 4.3** (Moy–Prasad depth as error rate, [\[my conjecture\]](#)). *Let  $\mathcal{C}$  be a quantum code with associated representation  $\pi_{\mathcal{C}}$  of a  $p$ -adic group (via the Langlands bridge of Pillar VII). Then the Moy–Prasad depth  $r(\pi_{\mathcal{C}})$  and the logical error rate  $p_L$  satisfy*

$$p_L = \exp(-r(\pi_{\mathcal{C}})/\tau)$$

*for a universal temperature-like parameter  $\tau > 0$  (possibly  $\tau = k_B T/E_0$ ). Representations of depth 0 (tamely ramified) correspond to fault-tolerant codes with exponential error suppression; positive depth codes require concatenation.*

*Remark 4.2* (Falsifiability of Conjecture 4.3). Compute Moy–Prasad depths for known code families where the  $p$ -adic representation is constructible (e.g., codes arising from algebraic constructions: Goppa codes, AG codes) and correlate with measured logical error rates at fixed physical error rates. No exponential relationship falsifies the depth–rate link.

# 5 The Fontaine Classification (Cross-Pillar)

## 5.1 The Code Filtration

**Conjecture 5.1** (Fontaine filtration of stabilizer codes, [\[my conjecture\]](#)). *There exists a nested classification of stabilizer codes over  $\mathbb{Q}_p$ :*

$$\mathcal{C}_{\text{crys}} \subset \mathcal{C}_{\text{st}} \subset \mathcal{C}_{\text{dR}} \subset \mathcal{C}_{\text{HT}} \subset \mathcal{C}_{\text{all}},$$

where each class corresponds to the analogous Fontaine period-ring condition on the Witt cohomology  $H_{\text{stab}}^\bullet(\mathcal{C}) \otimes_{\mathbb{Z}_p} \mathbb{Q}_p$ , viewed as a  $G_{\mathbb{Q}_p}$ -representation:

$$\mathcal{C} \in \mathcal{C}_\bullet \iff \dim_{\mathbb{Q}_p}(H_{\text{stab}}^\bullet(\mathcal{C}) \otimes_{\mathbb{Q}_p} B_\bullet)^{G_{\mathbb{Q}_p}} = \dim_{\mathbb{Q}_p} H_{\text{stab}}^\bullet(\mathcal{C}) \otimes \mathbb{Q}_p,$$

for  $B_\bullet \in \{B_{\text{crys}}, B_{\text{st}}, B_{\text{dR}}, B_{\text{HT}}\}$ .

*Remark 5.1* (Falsifiability of Conjecture 5.1). The conjecture requires constructing a  $G_{\mathbb{Q}_p}$ -action on  $H_{\text{stab}}^\bullet(\mathcal{C})$ . If no natural  $G_{\mathbb{Q}_p}$ -action can be defined, or if the filtration collapses ( $\mathcal{C}_{\text{crys}} = \mathcal{C}_{\text{dR}}$  for all known codes), the Fontaine analogy is falsified.

## 5.2 Interpretation of Fontaine Classes

**Proposition 5.1** (Fontaine class semantics, [my conjecture]).

| Code class                  | Number-theoretic analog    | QEC meaning                                                                                           |
|-----------------------------|----------------------------|-------------------------------------------------------------------------------------------------------|
| $\mathcal{C}_{\text{crys}}$ | Good reduction, unramified | <b>Rigid codes:</b> lift from $\mathbb{F}_p$ without deformation;<br>sharp fault-tolerance thresholds |
| $\mathcal{C}_{\text{st}}$   | Semistable reduction       | <b>Flexible codes:</b> controlled Clifford deformation;<br>monodromy encodes logical gates            |
| $\mathcal{C}_{\text{dR}}$   | All admissible reps        | <b>All analytic codes:</b> expressible in $p$ -adic Hodge framework                                   |
| $\mathcal{C}_{\text{HT}}$   | Hodge–Tate representations | Weighted-stabilizer codes: stratified by Hodge–Tate weights                                           |

## 5.3 The Frobenius–Cyclic Measurement Correspondence

**Conjecture 5.2** (Frobenius as cyclic measurement, [my conjecture]). The Frobenius endomorphism on the crystalline Dieudonné module

$$\varphi : D_{\text{crys}}(H_{\text{stab}}^\bullet(\mathcal{C}) \otimes \mathbb{Q}_p) \longrightarrow D_{\text{crys}}(H_{\text{stab}}^\bullet(\mathcal{C}) \otimes \mathbb{Q}_p)$$

corresponds to the cyclic measurement operator  $M_{\text{cyclic}}$  of the silent-radix synthesis. That is, the following square commutes:

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{M_{\text{cyclic}}} & \mathcal{C}' \\ \downarrow & & \downarrow \\ D_{\text{crys}}(\mathcal{C}) & \xrightarrow{\varphi} & D_{\text{crys}}(\mathcal{C}') \end{array}$$

where  $M_{\text{cyclic}}(\mathcal{C})$  is the code obtained by applying a full cycle of Fourier measurements to  $\mathcal{C}$ , and both  $\varphi$  and  $M_{\text{cyclic}}$  are semilinear automorphisms whose eigenvalues are the Dieudonné–Manin slopes.

*Remark 5.2* (Falsifiability of Conjecture 5.2). Compute the slope spectrum of a known code  $\mathcal{C}$  from the Dieudonné decomposition (via the Witt cohomology functor) and independently compute the eigenvalue spectrum of the cyclic measurement operator on  $\mathcal{C}$ . The two spectra must coincide. A mismatch for any testable code falsifies the correspondence.

## 5.4 The Crystalline–Topological Link

**Conjecture 5.3** (Crystalline  $\iff$  topological, [my conjecture]). *A quantum code is topological (its error-correction properties are invariant under local perturbations of the Hamiltonian) if and only if it is crystalline — that is, its Witt cohomology factors through  $B_{\text{crys}}$ :*

$$\mathcal{C} \text{ is topological} \iff \dim_{\mathbb{Q}_p} (H_{\text{stab}}^\bullet(\mathcal{C}) \otimes B_{\text{crys}})^{G_{\mathbb{Q}_p}} = \dim_{\mathbb{Q}_p} H_{\text{stab}}^\bullet(\mathcal{C}) \otimes \mathbb{Q}_p.$$

*Remark 5.3* (Falsifiability of Conjecture 5.3). Verify on independently classified topological codes (toric code, color code, Kitaev surface code) and non-topological codes (stabilizer codes with no topological order). A topological code that is not crystalline, or a non-topological crystalline code, falsifies the equivalence.

## 6 Computational Consequences

### 6.1 Immediately Testable Predictions

We enumerate predictions that can be verified with existing code data and computational resources (Phase 2 of the research plan).

#### T1. Slope-threshold correlation. [my conjecture]

$$\text{Hypothesis: } \text{corr}(\Delta(\mathcal{C}), p_{\text{th}}(\mathcal{C}))_{\mathcal{C} \in \text{Families}} < -0.7.$$

*Method:* Compute slopes for known code families (surface, color, toric, LDPC) via the conjectural Dieudonné decomposition; correlate with published threshold data.

*Falsification:*  $|\text{corr}| < 0.3$  after controlling for  $n, k, d$ .

#### T2. Crystalline detection. [my conjecture]

$$\text{Hypothesis: } \{\text{Surface codes, color codes, toric codes}\} \subseteq \mathcal{C}_{\text{crys}}.$$

*Method:* Test Frobenius-invariance of the stabilizer: verify  $USU^\dagger = S$  for full Clifford group elements  $U$ .

*Falsification:* Any topological code failing full Clifford invariance.

#### T3. Witt truncation order. [my conjecture]

$$\text{Hypothesis: } \mathcal{C}_n \text{ corrects errors up to weight } (n-1)/2 \text{ on the lifted stabilizer.}$$

*Method:* Construct explicit  $\mathcal{C}_n$  for concatenated codes and test error-correction capacity.

*Falsification:*  $\mathcal{C}_1$  outperforming  $\mathcal{C}_2$ , or  $\mathcal{C}_n$  for large  $n$  failing to improve.

#### T4. Adelic product constraint. [my conjecture]

$$\text{Hypothesis: } \prod_{p \leq \infty} d_p(\mathcal{C}) = 1,$$

where  $d_p(\mathcal{C})$  is the  $p$ -adic distance of the local code at prime  $p$ .

*Method:* Construct local codes for several primes; compute  $p$ -adic distances; verify the product.

*Falsification:*  $\prod_p d_p \neq 1$  for any consistently constructed local code family.

## 6.2 Predictions Requiring Deeper Theory

These predictions require advancing the mathematical framework of §§2–5 before testing.

- T5. Brauer–Manin obstruction.** [\[speculative\]](#) Some code parameters  $[[n, k, d]]_q$  have local realizations at every prime  $p$  and at  $p = \infty$  but no global code over  $\mathbb{C}$ . The obstruction lies in the Brauer group  $\text{Br}(X)/\text{Br}(\mathbb{Q})$  for a suitable moduli space  $X$  of codes.
- T6. Bernstein universality bound.** [\[speculative\]](#) The number of asymptotic universality classes of quantum code families (distinguished by their scaling behavior  $k \sim n^\alpha$ ,  $d \sim n^\beta$ ) is finite for each fixed qudit dimension, bounded by the number of Bernstein blocks for  $\text{GL}_n(\mathbb{Q}_p)$ .
- T7. Langlands  $\varepsilon$ -factor = MacWilliams transform.** [\[speculative\]](#) For a quantum code  $\mathcal{C}$  with weight enumerator  $A_{\mathcal{C}}(z)$ , let  $L(s, \pi_{\mathcal{C}})$  be the  $L$ -function of the associated Langlands parameter. Then the functional equation  $\varepsilon$ -factor satisfies

$$|\varepsilon(1/2, \pi_{\mathcal{C}})| = \frac{1}{q^{\kappa/2}} \cdot A_{\mathcal{C}}\left(\frac{-1}{q+1}\right).$$

- T8. Exceptional code symmetries.** [\[speculative\]](#) Codes corresponding to exceptional Kodaira types  $\text{II}^*$ ,  $\text{III}^*$ ,  $\text{IV}^*$  have exceptional Lie group symmetries  $E_8, E_7, E_6$  in their stabilizer groups. Such codes have not been constructed and may represent a fundamentally new class of quantum codes.

## 6.3 Falsifiability of the Entire Pillar VI Framework

The whole Witt cohomology program (Pillars VI and III) is falsified as a scientific framework if *any* of the following conditions hold:

- (i) **No cohomological structure:** No natural map from the set of stabilizer codes to modules over the Dieudonné ring can be constructed — the functor  $H_{\text{stab}}^\bullet$  is everywhere zero or fails functoriality.
- (ii) **No slope–threshold correlation:** After computing slopes for at least 10 code families, the Pearson correlation between  $\Delta(\mathcal{C})$  and  $p_{\text{th}}(\mathcal{C})$  is indistinguishable from zero ( $|r| < 0.2$ ).
- (iii) **No natural  $F, V$  operators:** The operations of Clifford conjugation and  $T$ -gate injection on stabilizer codes do not induce endomorphisms with the relation  $FV = p = VF$  on *any* code-theoretic invariant.
- (iv) **Fontaine filtration collapse:** The filtration  $\mathcal{C}_{\text{crys}} \subset \mathcal{C}_{\text{st}} \subset \mathcal{C}_{\text{dR}}$  collapses to a single class — i.e., all stabilizer codes are crystalline (or all are de Rham), providing no discrimination.
- (v) **Truncation fails to encode correction order:**  $\mathcal{C}_1$  and  $\mathcal{C}_n$  for  $n > 1$  have identical error-correction properties.

Conversely, confirmation of the slope–threshold correlation at  $|r| > 0.7$  would constitute strong evidence for the geometric picture underlying Pillar VI, even before the full functor is constructed.

## 6.4 Computational Implementation Roadmap

- (a) `dieudonne_slope_classifier.py`: Given a stabilizer code  $\mathcal{C}$ , compute its conjectural slope spectrum by analyzing the action of the Clifford group on the stabilizer. Output  $\text{Slopes}(\mathcal{C})$  and  $\Delta(\mathcal{C})$ .
- (b) `witt_truncation_builder.py`: Construct  $\mathcal{C}_n = \mathcal{C} \otimes W_n(\mathbb{F}_q)$  for small  $n$  and small codes. Test error correction order.
- (c) `fontaine_classifier.py`: Given  $\mathcal{C}$ , determine whether the Witt cohomology (as a  $G_{\mathbb{Q}_p}$ -representation) is crystalline, semistable, or de Rham by checking Frobenius-invariance and monodromy.
- (d) `falsifiability_dashboard.py`: Given a corpus of code families, compute correlation  $r(\Delta, p_{\text{th}})$  and report whether each falsifiability condition (i)–(v) of §6.3 is triggered.

## Acknowledgments

This document synthesizes conjectures from the NUMBER-THEORY-QEC-BRIDGE.MD (all seven pillars) and the RESEARCH-PLAN.MD (Pillars III and VI) of the `number-theory-ultrametric-deep` project. All non-textbook claims are labeled with certainty per QNFO-POL-COM-001. Each conjecture includes an explicit falsifiability condition.